

Other Software

Version 5.6 of AIM-Spice was released on April 15, 2012. Version 5.6 supports BSIM4v7.0.

Other similar software

LTspice IV. Produced by Linear Technology, LTspice IV is a high-performance SPICE simulator, waveform viewer and schematic capture tool with models and enhancements for easing the simulation of switching regulators. Compared to normal SPICE simulators, simulating switching regulators is extremely fast, allowing the user to view waveforms for most switching regulators in just a few minutes. Included with the software are LTspice IV, Macro Models for 80 per cent of Linear Technology's switching regulators, over 200 op-amp models, as well as transistor, resistor and MOSFET models.

CoolSpice. The CoolSpice package is a circuit simulation tool that includes a schematics editor with a plotter, GUI and text editor for editing netlists manually. The student version of CoolSpice is currently available for free. CoolSpice comes with two different Visual C++ Redistributables for 32- and 64-bit machines.

CoolSpice is CoolCAD's proprietary SPICE suite and circuit simulation software package.

SapWin. For the Windows (95/98/ME/NT/2000) environment, SAPWIN (Symbolic Analysis Program for Windows) is a proprietary symbolic circuit simulator written in C++. Unlike more common circuit simulators (such as SPICE), SAPWIN can generate analytical Laplace domain expressions for arbitrary network functions of linear analogue circuits. The SAPWIN package also includes tools for graphic post-processing and schematic capture. SAPWIN is available free.

Oregano (software) Linux only. Oregano is a graphical software application for schematic capture and simulation of electrical circuits. The actual simulation is performed by ngspice or GnuCap engines. It is similar to gEDA and KTeclab. It makes use of GNOME technology and is meant to run on open source Unix environments like Linux or FreeBSD.

TINA-TI. A circuit design and simulation tool, TINA-TI is ideal for testing, designing and troubleshooting a broad variety of basic and advanced circuits, without any number of device limitations or node limitations, including complex architectures. Teaming up with DesignSoft, TI has selected TINA simulation software over other SPICE-based simulators for its combination of simple and intuitive graphics-based interface, powerful analysis capabilities and ease of use. If one is familiar with any other SPICE simulator, adapting to TINA-TI should be an easy, straightforward transition. Although TINA-TI is a limited version, it easily handles surprisingly complex circuits.

Anasoft's SuperSpice. SuperSpice has been specifically designed for use as an analogue simulation tool by practicing analogue engineers, for analogue engineers. It is designed to run simulation, edit schematics, probe signals and view waveforms. All key issues have been automated. One can modify the schematic, rerun and inspect new waveforms without having to reload graphs.

5Spice. 5Spice provides Spice-specific schematic entry, integrated graphing of simulation results, and the ability to define and save an unlimited number of analyses. It also incorporates easy inclusion of Spice/PSpice models from a user-expandable library. At component level, the focus is on analogue circuit analysis and design.

5Spice has analysis capabilities needed by experienced circuit designers while remaining easy to use. For more powerful features, choose the new 5Spice Professional or standard 2.0 level.

versions of the unified charge control model, which describes the channel charge in field-effect transistors (FETs) in the above and below threshold regimes, using one analytical expression.

Extensions to Berkeley SPICE

1. Temperature sweep analysis
2. Support for libraries
3. Libraries consisting of thousands of devices
4. Support for parameters in the netlist
5. Support for parameters in sub-circuit calls
6. Polynomial dependent sources (POLY sources)
7. Global nodes
8. BSIM3 versions 2 and 3
9. BSIM4 versions 1, 2, 2.1, 3 and 4
10. Support for extended commands:
 - .connect
 - .defwave
 - .global
 - .lib
 - .param
 - .plot
11. Enhanced commenting features (comment blocks and comments within continuation lines)

AIM-Postprocessor

As mentioned earlier, a graphic post-processor is included in the AIM-Spice simulator package. Like AIM-Spice, this application also runs under the Microsoft Windows environment.

Although AIM-Spice has facilities to plot circuit variables graphically, the postprocessor has a much more powerful plotting engine including the following features:

1. Creating hard copies and plotting of derivatives, integrals, sums and differences, and mathematical functions of circuit variables
2. FFT (fast Fourier transform)
3. Calculation of differences between variables and cursors to select numerical values
4. Import of experimental data
5. Hard copies ●

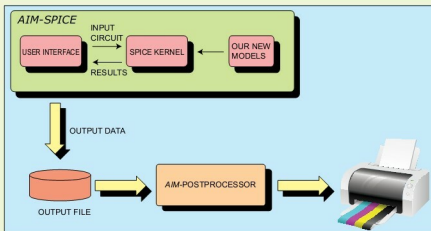


Fig. 4: Overview of the AIM-Spice simulator package

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